Jens Einar Bremnes

PhD · Marine Cybernetics

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Research Interests

My research addresses autonomy for robotics, combining risk analysis, AI, optimization and control theory to enable safe and reliable autonomous operations in real-world environments. Core topics include:

- Autonomy algorithms for unmanned systems, including mission planning, path planning and decision-making
- Probabilistic AI and machine learning (Gaussian processes, Bayesian methods) for risk awareness
- Sensor fusion, observer design, and underwater navigation
- Field robotics, including Arctic and under-ice environments

Education

Ph.D. in Marine Technology — Marine Cybernetics *2019–2024*
Norwegian University of Science and Technology (NTNU), Trondheim, Norway

Thesis: *Safe Autonomy in Marine Robotics*

PhD candidate in the UNLOCK project at the Dept. of Marine Technology. Affiliated with the Centre of Excellence for Autonomous Marine Operations and Systems (SFF AMOS) and the Nansen Legacy project. My research focused on risk modelling, risk control and autonomous planning and decision-making for marine robotics operating in unstructured environments.

M.Sc. in Marine Technology — Marine Cybernetics *2014–2019*
Norwegian University of Science and Technology (NTNU), Trondheim, Norway

Thesis: *Towards Robust Autonomy of Underwater Vehicles in Arctic Operations*

Grade average: 4.3 (B+). Coursework in advanced control systems, robotics, risk analysis, artificial intelligence, hydrodynamics, and programming. Began an integrated PhD programme in the fifth year.

Exchange studies:

- **Spring 2020** — Dept. of EECS, *UC Berkeley*, USA. Reachability analysis, Gaussian processes for machine learning, and model predictive control for robotics.
- **Fall 2019** — *UNIS*, Longyearbyen, Svalbard. Doctoral course *AT-834 Arctic Marine Measurement Techniques, Operations and Transport*.
- **Spring 2018** — Dept. of ECE, *Instituto Superior Técnico (IST)*, Lisbon. Robotics, optimisation, control theory, offshore wind energy.

Professional Experience

Engineer *Sep 2025–present*

GeoNord AS, Alta, Norway

Seabed and terrain mapping: field data collection with laser scanners and multibeam echo sounders from drone and vessel; processing and analysis of point clouds, 3D models, orthophotos, contour maps, and volume calculations.

Researcher *May 2023–Sep 2025*

Norwegian Defence Research Institute (FFI), Kjeller, Norway

Research and development on autonomy for HUGIN-class AUVs. Developed novel algorithms for autonomous mission planning and decision-making; implemented in C++ (primary), Python, and MATLAB with integration on operational AUV systems and operator interfaces. Regular field campaigns aboard research vessel *H.U. Sverdrup II*. Scientific dissemination via articles, conference papers, and reports. Supervised master's students and doctoral candidates. Contributed to research projects and project proposals.

Research Fellow (Ph.D.) *Aug 2019–May 2023*

Department of Marine Technology, NTNU, Trondheim, Norway

Autonomy research in marine robotics with focus on risk management, AI, and control systems. Contributed to the Applied Underwater Robotics Laboratory (AUR-Lab): operation and maintenance of AUV, USV, and ROV fleets; scientific cruises in the North Sea, Svalbard, Lake Mjøsa, and Trøndelag fjords; equipment testing and student training. Co-developed *TMR06 Autonomous Marine Systems* (master's course, first taught Fall 2020); teaching assistant for *TMR4240 Marine Control Systems*; co-supervised master's students.

Summer Intern — Digital Assurance Division *Summer 2018*

DNV GL, Trondheim, Norway

Software development in C++ and Python with ROS for the *ReVolt* autonomous ship testbed; vessel maintenance.

Summer Intern — New Energy Solutions *Summer 2017*

Equinor, Stavanger, Norway

Monopile design and parametric studies for large-scale (10+ MW) offshore wind turbines; structural analysis in ULS and FLS.

Research Projects

UNLOCK *2018–2023*

Research Council of Norway, FRINATEK (project no. 274441) / NTNU — PhD Candidate

Unlocking the Potential of Autonomous Systems and Operations through Supervisory Risk Control. The project involved online risk modelling, testing, and verification of safe autonomous systems, with case studies on AUVs for ocean mapping and UAVs for industrial inspection. My work focused on supervisory risk control and safe autonomy for marine robotics. Hosted at the Dept. of Marine Technology, NTNU; affiliated with SFF AMOS.

The Nansen Legacy *2021*

Research Council of Norway / UiT – The Arctic University of Norway — Participant

Large-scale interdisciplinary research programme on the Arctic marine ecosystem and its future. Participated in the Q2 2021 seasonal cruise in the Arctic Ocean, contributing to ROV-based data collection, under-ice operations, and bio-optical measurements.

SmartAUVs *2022–2026*

Research Council of Norway / Plymouth Marine Laboratory — Collaborator (FFI)

Smart AUVs for Detection and Quantification of Greenhouse Gas Seepage in the Oceans. Developed autonomy algorithms for adaptive sampling of CO_2 and methane seepage triggered by real-time detections.

Intelligent ID *2023–2025*

Norwegian Defence Research Institute (FFI) — Lead developer

FFI project on autonomous optical object identification using AUVs. Developed replanning algorithms for robust in-situ sonar-guided optical inspection with the HUGIN AUV.

AutoCSAS *2023–2025*

Norwegian Defence Research Institute (FFI) — Lead developer

FFI project on decisional autonomy for circular synthetic aperture sonar (CSAS) surveying. Developed autonomous mission planning and replanning strategies for AUV-based CSAS operations.

Publications

Citations: **129** h-index: **7**

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Peer-Reviewed Journal Articles

[1] N. Summers, **J. E. Bremnes**, H. Liu, H. S. Løvås, G. M. Fragoso, and G. Johnsen, "Assessing Arctic under sea-ice light regimes and ice algal bio-optical properties using ROV-based hyperspectral imaging," *Polar Biology*, vol. 49, pp. 1–26, 2026. [doi]

[2] S. D. Kristensen, R. G. Maidana, I. B. Utne, and **J. E. Bremnes**, "Evaluating the effect of risk metrics for supporting operational decision-making by autonomous surface vehicles," *Ocean Engineering*, vol. 338, p. 121937, 2025. [doi]

[3] **J. E. Bremnes**, I. B. Utne, T. R. Krogstad, and A. J. Sørensen, "Holistic risk modeling and path planning for marine robotics," *IEEE Journal of Oceanic Engineering*, vol. 50, no. 1, pp. 252–275, 2024. [doi]

[4] **J. E. Bremnes**, T. R. Fyrvik, T. R. Krogstad, and A. J. Sørensen, "Design of a switching controller for tracking AUVs with an ASV," *IEEE Transactions on Control Systems Technology*, vol. 32, no. 5, pp. 1785–1800, 2024. [doi]

[5] A. Vasilijevic, **J. E. Bremnes**, and M. Ludvigsen, "Remote operation of marine robotic systems and next-generation multi-purpose control rooms," *Journal of Marine Science and Engineering*, vol. 11, no. 10, pp. 1–25, 2023. [doi]

[6] R. Yang, **J. E. Bremnes**, and I. B. Utne, "Online risk modeling of autonomous marine systems: Case study of autonomous operations under sea ice," *Ocean Engineering*, vol. 281, p. 114765, 2023. [doi]

[7] **J. E. Bremnes**, A. H. Brodtkorb, and A. J. Sørensen, "Hybrid observer concept for sensor fusion of sporadic measurements for underwater navigation," *International Journal of Control, Automation and Systems*, vol. 19, pp. 137–144, 2020. [doi]

[8] **J. E. Bremnes**, C. A. Thieme, A. J. Sørensen, I. B. Utne, and P. Norgren, "A Bayesian approach to supervisory risk control of AUVs applied to under-ice operations," *Marine Technology Society Journal*, vol. 54, no. 4, pp. 16–39, 2020. [doi]

Conference Proceedings

[1] **J. E. Bremnes** and T. R. Krogstad, "Planning in circles: Autonomous planning for circular synthetic aperture sonar surveying with AUVs," in *Proc. 2025 Undersea Defence Technology Conference (UDT)*, Oslo, Norway, 2025, pp. 1–6.

[2] **J. E. Bremnes**, H. Y. Rørstad, T. R. Krogstad, and Ø. Midtgaard, "Adaptive AUV planning for robust optical object inspection using in-situ sonar data," in *Proc. 2024 IEEE/OES Autonomous Underwater Vehicle Symposium (AUV)*, Boston, MA, USA, 2024, pp. 1–6. [doi]

[3] A. Vasilijevic, H. Gravråkmo, K. Barstein, and **J. E. Bremnes**, "Infrastructure for remote experimentation in the Trondheim Fjord," in *Proc. OCEANS 2023*, Limerick, Ireland, 2023, pp. 1–6. [doi]

[4] R. Yang, **J. E. Bremnes**, and I. Utne, "A system-theoretic approach to hazard identification of operation with multiple autonomous marine systems (AMS)," in *Proc. 32nd European Safety and Reliability Conference (ESREL)*, Dublin, Ireland, 2022, pp. 1–8. [doi]

[5] T. R. Fyrvik, **J. E. Bremnes**, and A. J. Sørensen, "Hybrid tracking controller for an ASV providing mission support for an AUV," in *Proc. 14th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles (CAMS)*, Copenhagen, Denmark, 2022, pp. 1–7. [doi]

[6] **J. E. Bremnes**, A. Devonport, H. Yin, M. Arcak, A. J. Sørensen, and I. B. Utne, "Optimization-based planning and control of AUVs applied to adaptive sampling under ice," in *Proc. 2020 IEEE/OES Autonomous Underwater Vehicle Symposium (AUV)*, St. Johns, Canada, 2020, pp. 1–6. [doi]

[7] **J. E. Bremnes**, P. Norgren, A. J. Sørensen, C. A. Thieme, and I. B. Utne, "Intelligent risk-based under-ice altitude control for autonomous underwater vehicles," in *Proc. IEEE/MTS OCEANS 2019 Conference*, Seattle, WA, USA, 2019, pp. 1–8. [doi]

[8] **J. E. Bremnes**, A. H. Brodtkorb, and A. J. Sørensen, "Sensor-based hybrid translational observer for underwater navigation," in *Proc. 12th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles (CAMS)*, Daejeon, South Korea, 2019, pp. 378–383. [doi]

Theses

[1] **J. E. Bremnes**, *Safe Autonomy in Marine Robotics*, PhD thesis, Dept. of Marine Technology, Norwegian University of Science and Technology (NTNU), Trondheim, Norway, 2024.

[2] **J. E. Bremnes**, *Towards Robust Autonomy of Underwater Vehicles in Arctic Operations*, Master's thesis, Dept. of Marine Technology, NTNU, Trondheim, Norway, 2019.

Reports

[1] **J. E. Bremnes** and T. R. Krogstad, "CsasPlanner: Decisional autonomy for circular synthetic aperture sonar surveying with AUVs," *FFI Report*, Norwegian Defence Research Institute (FFI), 2025. *[Exempt from public access.]*

[2] **J. E. Bremnes** et al., "Intelligent ID: Replanning for robust optical identification with HUGIN using EM2040," *FFI Report*, Norwegian Defence Research Institute (FFI), 2025. *[Exempt from public access.]*

[3] M. Ludvigsen et al. (incl. **J. E. Bremnes**), "Seasonal cruise Q2 2021: Cruise Report," *The Nansen Legacy Report Series*, 2022. [doi]

[4] D. Metlay, A. L. St. Clair, B. Twomey, **J. E. Bremnes**, Ø. J. Rødseth, and S. Massaiu, "Autonomous transportation technologies: Society and individuals in the loop. Report of the discussions of breakout session," in *Proc. 1st International Workshop on Autonomous Systems Safety*, 2019.

Teaching & Supervision

- **Course co-developer** — *TMR06 Autonomous Marine Systems*, NTNU (first taught Fall 2020). Contributed to syllabus design, lecture notes, and project specification.
- **Teaching assistant** — *TMR4240 Marine Control Systems*, NTNU (2019–2023). Exercise supervision, grading, and student guidance.
- **Teaching assistant** — *TMR4160 Data Methods for Marine Applications*, NTNU.
- **Co-supervision** of master's students (NTNU) and doctoral candidates (FFI and NTNU).
- **Supply teacher**, mathematics and natural science, grades 8–10, Sandfallet Secondary School,

Alta (Jan 2016, 2017, 2018).

- **Science mentor**, ENT3R, Trondheim (2015–2017) — weekly tutoring and hands-on science experiments for secondary school students.

Outreach & Science Communication

- **J. E. Bremnes** et al. (2021). "Leveraging ocean data harvesting by heterogeneous robotic organisations and autonomous vehicles as sensor-carrying platforms." *Breaking the Surface (BtS)*, Croatia. *[Invited lecture]*
- L. Aarønæs (2024). "Han vil lære undervannsroboter å frykte døden" [He wants to teach underwater robots to fear death]. *ffi.no*. [link]
- N. Summers, **J. E. Bremnes** et al. (2021). "Mixing production deep into the ocean." *sciencenorway.no*. [link]
- N. Summers, **J. E. Bremnes** et al. (2021). "Våren ligger i lufta." *forskning.no*. [link]

Technical Skills

Programming	C++, Python, MATLAB / Simulink
Frameworks & tools	ROS (Robot Operating System), Git, Docker, Linux, GIS
Methods	Control theory, sensor fusion, Gaussian processes, Bayesian inference, risk analysis, autonomous planning, data visualisation